

Table 11.2 Formulae for the Likelihood Ratio for Situations Where the Genotype of One Parent of the Missing Person is Available as a Reference Sample

Parent	Body	LR Assessed By the Product Rule	LR Assessed Using the Subpopulation Correction (mother, parent and body in the same subpopulation)
<i>aa</i>	<i>aa</i>	$\frac{1}{p_a}$	$\frac{1 + 2\theta}{3\theta + (1 - \theta)p_a}$
	<i>ab</i>	$\frac{1}{2p_a}$	$\frac{1 + 2\theta}{2(2\theta + (1 - \theta)p_a)}$
<i>ab</i>	<i>aa</i>	$\frac{1}{2p_a}$	$\frac{1 + 2\theta}{2(2\theta + (1 - \theta)p_a)}$
	<i>ab</i>	$\frac{p_a + p_b}{4p_a p_b}$	$\frac{(1 + 2\theta)[2\theta + (1 - \theta)(p_a + p_b)]}{4(\theta + (1 - \theta)p_a)(\theta + (1 - \theta)p_b)}$
	<i>ac</i>	$\frac{1}{4p_a}$	$\frac{1 + 2\theta}{4(\theta + (1 - \theta)p_a)}$

Table 11.4 Formulae for the Likelihood Ratio for Situations Where the Genotype of One Child of the Missing Person is Available as a Reference Sample But the Partner is Unavailable

Body	Child	<i>LR</i> Assessed By the Product Rule	<i>LR</i> Assessed Using the Subpopulation Correction (body and child members of the same subpopulation)
<i>aa</i>	<i>aa</i>	$\frac{1}{p_a}$	$\frac{1 + 2\theta}{3\theta + (1 - \theta)p_a}$
	<i>ab</i>	$\frac{1}{2p_a}$	$\frac{1 + 2\theta}{2[2\theta + (1 - \theta)p_a]}$
<i>ab</i>	<i>aa</i>	$\frac{1}{2p_a}$	$\frac{1 + 2\theta}{2[2\theta + (1 - \theta)p_a]}$
	<i>ab</i>	$\frac{p_a + p_b}{4p_a p_b}$	$\frac{(1 + 2\theta)[2\theta + (1 - \theta)(p_a + p_b)]}{4(\theta + (1 - \theta)p_a)(\theta + (1 - \theta)p_b)}$
	<i>ac</i>	$\frac{1}{4p_a}$	$\frac{1 + 2\theta}{4[\theta + (1 - \theta)p_a]}$

Table 11.5 Formulae for the Likelihood Ratio for Situations Where the Genotype of One Sibling of the Missing Person is Available as a Reference Sample

Sibling	Body		<i>LR</i>
		Product Rule	Using Subpopulation Correction (sibling and body in the same subpopulation)
<i>aa</i>	<i>aa</i>	$\frac{(1+p_a)^2}{4p_a^2}$	$\frac{(1+\theta)(1+2\theta)+2(1+2\theta)(2\theta+(1-\theta)p_a)+(2\theta+(1-\theta)p_a)(3\theta+(1-\theta)p_a)}{4(2\theta+(1-\theta)p_a)(3\theta+(1-\theta)p_a)}$
	<i>bb</i>	$\frac{1}{4}$	$\frac{1}{4}$
	<i>ab</i>	$\frac{1+p_a}{4p_a}$	$\frac{1+4\theta+(1-\theta)p_a}{4(2\theta+(1-\theta)p_a)}$
	<i>bc</i>	$\frac{1}{4}$	$\frac{1}{4}$
<i>ab</i>	<i>aa</i>	$\frac{1+p_a}{4p_a}$	$\frac{1+4\theta+(1-\theta)p_a}{4(2\theta+(1-\theta)p_a)}$
	<i>ab</i>	$\frac{1+p_a+p_b+2p_ap_b}{8p_ap_b}$	$\frac{(1+\theta)(1+2\theta)+(1+2\theta)(2\theta+(1-\theta)(p_a+p_b))+2(\theta+(1-\theta)p_a)(\theta+(1-\theta)p_b)}{8(\theta+(1-\theta)p_a)(\theta+(1-\theta)p_b)}$
	<i>ac</i>	$\frac{1+2p_a}{8p_a}$	$\frac{1+4\theta+2(1-\theta)p_a}{8(\theta+(1-\theta)p_a)}$
	<i>cc</i>	$\frac{1}{4}$	$\frac{1}{4}$
	<i>cd</i>	$\frac{1}{4}$	$\frac{1}{4}$

Table 11.3 Formulae for the Likelihood Ratio for Situations Where the Genotypes of the Partner and One Child of the Missing Person are Available as Reference Samples

Body	Child	Partner of the Missing Person	LR Assessed By the Product Rule	LR Assessed Using the Subpopulation Correction (partner, child, and body in the same subpopulation)
<i>aa</i>	<i>aa</i>	<i>aa</i>	$\frac{1}{p_a}$	$\frac{1 + 3\theta}{4\theta + (1 - \theta)p_a}$
		<i>ab</i>	$\frac{1}{p_a}$	$\frac{1 + 3\theta}{3\theta + (1 - \theta)p_a}$
	<i>ab</i>	<i>ab</i>	$\frac{1}{p_a + p_b}$	$\frac{1 + 3\theta}{4\theta + (1 - \theta)(p_a + p_b)}$
		<i>bb</i> or <i>bc</i>	$\frac{1}{p_a}$	$\frac{1 + 3\theta}{2\theta + (1 - \theta)p_a}$
<i>ab</i>	<i>aa</i>	<i>aa</i>	$\frac{1}{2p_a}$	$\frac{1 + 3\theta}{2[3\theta + (1 - \theta)p_a]}$
		<i>ab</i> or <i>ac</i>	$\frac{1}{2p_a}$	$\frac{1 + 3\theta}{2[2\theta + (1 - \theta)p_a]}$
	<i>ab</i>	<i>ab</i>	$\frac{1}{p_a + p_b}$	$\frac{1 + 3\theta}{4\theta + (1 - \theta)(p_a + p_b)}$
		<i>bb</i> or <i>bc</i>	$\frac{1}{2p_a}$	$\frac{1 + 3\theta}{2[\theta + (1 - \theta)p_a]}$
	<i>ac</i>	<i>cc</i> or <i>cd</i>	$\frac{1}{2p_a}$	$\frac{1 + 3\theta}{2[\theta + (1 - \theta)p_a]}$

Table 11.1 Formulae for the Likelihood Ratio for Situations Where Genotypes of Both Parents (But No Children) of the Missing Person are Available as Reference Samples

Parent 1	Parent 2	Body	LR Assessed By the Product Rule	LR Assessed Using the Subpopulation Correction (both parents and the body are members of the same subpopulation)
<i>aa</i>	<i>aa</i>	<i>aa</i>	$\frac{1}{p_a^2}$	$\frac{(1 + 3\theta)(1 + 4\theta)}{(4\theta + (1 - \theta)p_a)(5\theta + (1 - \theta)p_a)}$
	<i>ab</i>	<i>aa</i>	$\frac{1}{2p_a^2}$	$\frac{(1 + 3\theta)(1 + 4\theta)}{2(3\theta + (1 - \theta)p_a)(4\theta + (1 - \theta)p_a)}$
		<i>ab</i>	$\frac{1}{4p_a p_b}$	$\frac{(1 + 3\theta)(1 + 4\theta)}{4(3\theta + (1 - \theta)p_a)(\theta + (1 - \theta)p_b)}$
		<i>bb</i>	$\frac{1}{2p_a p_b}$	$\frac{(1 + 3\theta)(1 + 4\theta)}{2(2\theta + (1 - \theta)p_a)(2\theta + (1 - \theta)p_b)}$
	<i>bc</i>	<i>ab</i>	$\frac{1}{4p_a p_b}$	$\frac{(1 + 3\theta)(1 + 4\theta)}{4(2\theta + (1 - \theta)p_a)(\theta + (1 - \theta)p_b)}$
<i>ab</i>	<i>ab</i>	<i>aa</i>	$\frac{1}{4p_a^2}$	$\frac{(1 + 3\theta)(1 + 4\theta)}{4(2\theta + (1 - \theta)p_a)(3\theta + (1 - \theta)p_a)}$
		<i>ab</i>	$\frac{1}{4p_a p_b}$	$\frac{(1 + 3\theta)(1 + 4\theta)}{4(2\theta + (1 - \theta)p_a)(2\theta + (1 - \theta)p_b)}$
	<i>ac</i>	<i>ab</i>	$\frac{1}{8p_a p_b}$	$\frac{(1 + 3\theta)(1 + 4\theta)}{8(2\theta + (1 - \theta)p_a)(\theta + (1 - \theta)p_b)}$
		<i>aa</i>	$\frac{1}{4p_a^2}$	$\frac{(1 + 3\theta)(1 + 4\theta)}{4(2\theta + (1 - \theta)p_a)(3\theta + (1 - \theta)p_a)}$
	<i>cd</i>	<i>ac</i>	$\frac{1}{8p_a p_c}$	$\frac{(1 + 3\theta)(1 + 4\theta)}{8(\theta + (1 - \theta)p_a)(\theta + (1 - \theta)p_c)}$